

# PEM-CAN: Parameter-Efficient Multimodal Diabetic Neuropathy Screening

Orthogonal Low-Rank Subspace Cross-Attention over Retinal Imaging & Wearable Biosignals

## 1. MULTIMODAL SENSORS



### Retinal Oculomics

Cumulative Microvascular State

- High-resolution fundus photography
- Arteriolar narrowing & capillary rarefaction
- Spatial tokens:  $M = 576$



### Wearable Telemetry

Dynamic Autonomic Excursions

- 7-day CGM (5-min,  $T = 2,016$  steps)
- 5-channel actigraphy: HRV, sleep, stress
- Multiscale Conv1D tokens:  $L = 252$

### Clinical Challenge

Visual dominance & quadratic attention blowup

## 2. PEM-CAN SUBSPACE ADAPTERS



### Frozen ViT-B/16

86.6M Base Params  
(Zero Backprop)



### Temporal Conv

Multiscale Tokenizer  
( $L = 252$  steps)



### Stiefel LoRA

$\|AA^T - I_r\|_F^2$  Penalty  
Rank  $r = 8$  Subspace  
98.3k Adapter Weights

### Bidirectional Cross-Attn

$S(Z_v, Z_s)$  &  $S(Z_s, Z_v)$

- Linear Complexity:  $O(M \cdot L)$
- Prevents Modality Collapse
- Zero-Overhead Inference
- Adapters fold into  $W_0$

$L_{total} = L_{focal} + L_{orth} + L_{align}$

### Multi-Task Head

DAN (Autonomic) | DR (Retina)  
CKD (Renal) | DPN (Sensory)

## 3. BENCHMARK COHORT



### AI-READI T2D Protocol

$N = 2,840$  Patients

- Statistically mapped to NIH atlas
- All confirmed Type 2 Diabetes
- Stratified 5-Fold Cross-Validation
- White (60%), Black (25%), Lat (15%)



### Adjudicated Targets

Clinically Validated Labels

- DAN (Autonomic): 22.5% Prev.
- DR (Retinopathy): 18.2% Prev.
- CKD (Renal): 14.1% Prev.
- DPN (Peripheral): 26.8% Prev.

## 4. QUANTIFIED PERFORMANCE



### 0.934 AUROC

90.5% Accuracy | 0.803 F1

- Oracle Ceiling: 0.957
- ECE: 0.038



### 1.75% Weights (2.6M)

8.4 GB VRAM (vs. 32.4 GB)

- 0.861 AUROC under 50% sensor loss



### 20.2 ms Edge Latency

NVIDIA Jetson Orin Nano (1.18 GB)

- 50-Node FL: 98.25% less bandwidth